PYTHON OPERATORS MASTERY EXAMINATION

Topic Focus: Arithmetic, Relational, Logical, Assignment, Bitwise, Membership & Identity

Candidate Name: _____ **Date:** _____ **Total Questions: 50 | Score: ___/50**

SECTION A: ARITHMETIC OPERATORS (+, -, *, /, //, %, **)

Q01

```
a = 15 // 4
b = 15 % 4
print(a, b)
```

Output:

Q02

```
x = -15 // 4
y = -15 % 4
print(x, y)
```

Output:

Q03

```
res = 2 ** 3 ** 2
print(res)
```

Output:

Q04

```
p = 10 + 3 * 2 ** 2 - 4 / 2
print(p)
```

Output:

Q05

```
val = 10.0 % 3
print(type(val), val)
```

Output:

Q06

```
a = 7
b = 2.0
print(a // b)
```

Output:

Q07

```
x = 5.5 + 4.5 * 2 // 3
print(x)
```

Output:

SECTION B: RELATIONAL / COMPARISON OPERATORS (==, !=, >, <, >=, <=)

Q08

```
print(10 == 10.0)
```

Output:

Q09

```
print("apple" > "Apple")
```

Output:

Q10

```
print([1, 2] < [1, 2, 0])
```

Output:

Q11

```
print(True == 1, False == 0)
```

Output:

Q12

```
a = 5
print(1 < a < 10)
```

Output:

Q13

```
print(10 != 10.0)
```

Output:

SECTION C: LOGICAL OPERATORS (AND, OR, NOT)

Q14

```
x = 0 and 10
y = 5 and 20
print(x, y)
```

Output:

Q15

```
a = "Hello" or ""
b = "" or "World"
print(a, b)
```

Output:

Q16

```
print(not [] and not "0")
```

Output:

Q17

```
res = 10 or 1 / 0
print(res)
```

Output:

Q18

```
x = [] or [0] or False
print(x)
```

Output:

Q19

```
print(not (5 > 2 and 3 < 1))
```

Output:

Q20

```
a = True
b = False
print(a and not b or b)
```

Output:

SECTION D: ASSIGNMENT & COMPOUND OPERATORS (=, +=, -=, *=, /=, %=, //=, **=, :=)

Q21

```
a = 10
a += 5 * 2
print(a)
```

Output:

Q22

```
x = 8
x //= 3
print(x)
```

Output:

Q23

```
lst = [1, 2]
lst += [3, 4]
print(lst)
```

Output:

Q24

```
a = 5
a **= 2
print(a)
```

Output:

Q25

```
if (n := len("Python")) > 5:
    print(n)
```

Output:

Q26

```
x = 12
x %= 5
print(x)
```

Output:

SECTION E: BITWISE OPERATORS (&, |, ^, ~, <<, >>)

Q27

```
a = 5  # 0101
b = 3  # 0011
print(a & b, a | b)
```

Output:

Q28

```
a = 5  # 0101
b = 3  # 0011
print(a ^ b)
```

Output:

Q29

```
x = 10
print(~x)
```

Output:

Q30

```
print(~(-5))
```

Output:

Q31

```
x = 3  # 0011
print(x << 2)
```

Output:

Q32

```
x = 16 # 10000
print(x >> 3)
```

Output:

Q33

```
a = 1
print((a << 3) - 1)
```

Output:

SECTION F: MEMBERSHIP OPERATORS (IN, NOT IN)

Q34

```
s = "py" in "python"
print(s)
```

Output:

Q35

```
d = {"a": 1, "b": 2}
print(1 in d, "a" in d)
```

Output:

Q36

```
print(1 not in [1, 2, 3])
```

Output:

Q37

```
t = (1, [2, 3])
print(2 in t)
```

Output:

Q38

```
print("th" not in "python")
```

Output:

Q39

```
st = {1, 2, 3}
print(3 in st)
```

Output:

SECTION G: IDENTITY OPERATORS (IS, IS NOT)

Q40

```
a = [1, 2]
b = [1, 2]
print(a is b, a == b)
```

Output:

Q41

```
x = 256
y = 256
print(x is y)
```

Output:

Q42

```
m = "hello"
n = "hello"
print(m is n)
```

Output:

Q43

```
a = None
print(a is None)
```

Output:

Q44

```
p = [10]
q = p
print(p is not q)
```

Output:

Q45

```
x = True
y = 1
print(x is y)
```

Output:

SECTION H: COMBINED & SPECIAL OPERATOR PRECEDENCE CHALLENGES

Q46

```
a = 10
b = 20
c = 30
print(a < b and b < c or a > c)
```

Output:

Q47

```
x = 5
print(x & 1 == 0)
```

Output:

Q48

```
a = [1, 2]
b = a
b += [3]
print(a is b, a)
```

Output:

Q49

```
a = [1, 2]
b = a
b = b + [3]
print(a is b, a)
```

Output:

Q50

```
x = 10
y = 5
z = x > y and x & y or x ^ y
print(z)
```

Output:
